

**WENO CHUUK, Micronesia, Federated States of**WMO: **913340**Lat: **7.454N**Lon: **151.843E**Elev: **11**StdP: **14.69**Time Zone: **10.00 (E10)**Period: **94-19**WBAN: **40505****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |       |      |      |       |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-------|------|------|-------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |       |      | 99%  |       |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR    | MCDB | DP   | HR    | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)   | (f)  | (g)  | (h)   | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 8             | 75.4       | 76.1 | 69.5                          | 108.6 | 81.8 | 70.6 | 112.9 | 81.8 | 23.1                  | 80.5 | 20.8 | 81.2 | 7.0                   | 60   | 0.453 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 7.5                    | 89.7            | 81.3 | 89.0 | 81.0 | 88.2 | 80.6 | 84.3                | 87.9 | 82.9 | 87.1 | 82.0 | 86.5 | 8.9                  | 40   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 83.2                            | 173.7 | 87.3 | 81.8 | 165.7 | 86.4 | 80.8 | 159.9 | 85.7 | 48.3          | 88.0 | 46.9 | 87.2 | 46.0 | 86.8 | 88.9              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |             |             | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |             |             |             |              |             |              |    |    |
|-------------------|-------------|-------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|-------------|-------------|-------------|--------------|-------------|--------------|----|----|
|                   |             |             | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years  |             | n=20 years  |              | n=50 years  |              |    |    |
| 1%                | 2.5%        | 5%          | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min         | Max         | Min         | Max          | Min         | Max          | DB | WB |
| (a)               | (b)         | (c)         | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)         | (l)         | (m)         | (n)          | (o)         | (p)          |    |    |
| (4) <b>19.0</b>   | <b>17.5</b> | <b>15.9</b> | <b>73.3</b>                | <b>93.9</b> | <b>0.7</b>         | <b>4.3</b> | <b>72.9</b>                                        | <b>97.0</b> | <b>72.5</b> | <b>99.5</b> | <b>72.1</b> | <b>102.0</b> | <b>71.6</b> | <b>105.1</b> |    |    |
| (5)               |             |             | <b>69.8</b>                | <b>84.6</b> | <b>1.6</b>         | <b>1.9</b> | <b>68.6</b>                                        | <b>86.0</b> | <b>67.7</b> | <b>87.1</b> | <b>66.8</b> | <b>88.2</b>  | <b>65.6</b> | <b>89.6</b>  |    |    |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 82.7   | 82.6  | 82.4  | 82.8  | 83.1  | 83.0  | 82.7  | 82.5  | 82.4  | 82.4  | 82.7  | 82.8  | 82.8 |
|                                                                                  | DBStd       | 1.60   | 1.52  | 1.55  | 1.60  | 1.66  | 1.63  | 1.63  | 1.62  | 1.67  | 1.68  | 1.52  | 1.52  | 1.44 |
|                                                                                  | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDD50       | 11926  | 1010  | 907   | 1016  | 992   | 1023  | 981   | 1007  | 1003  | 973   | 1013  | 985   | 1016 |
|                                                                                  | CDD65       | 6451   | 545   | 487   | 551   | 542   | 558   | 531   | 542   | 538   | 523   | 548   | 535   | 551  |
|                                                                                  | CDH74       | 73624  | 6188  | 5459  | 6251  | 6301  | 6484  | 6086  | 6121  | 5995  | 5888  | 6235  | 6202  | 6414 |
|                                                                                  | CDH80       | 23768  | 1871  | 1605  | 1959  | 2121  | 2206  | 1999  | 1962  | 1892  | 1897  | 2082  | 2078  | 2096 |
| Wind                                                                             | WSAvg       | 8.7    | 10.8  | 11.1  | 10.5  | 9.5   | 8.2   | 7.7   | 7.1   | 7.4   | 7.6   | 7.2   | 7.8   | 9.6  |
| Precipitation                                                                    | PrecAvg     | 140.0  | 9.1   | 7.6   | 9.6   | 11.5  | 12.9  | 12.5  | 13.7  | 13.6  | 13.3  | 12.8  | 11.2  | 11.4 |
|                                                                                  | PrecMax     | 179.1  | 19.2  | 35.7  | 24.0  | 30.6  | 28.4  | 25.4  | 34.3  | 25.9  | 23.7  | 24.7  | 26.1  | 24.5 |
|                                                                                  | PrecMin     | 86.7   | 1.2   | 0.6   | 0.8   | 1.7   | 3.8   | 5.7   | 2.6   | 3.2   | 5.2   | 4.2   | 1.9   | 2.9  |
|                                                                                  | PrecStd     | 19.4   | 4.2   | 5.5   | 5.2   | 5.6   | 4.4   | 3.9   | 5.8   | 4.6   | 4.3   | 4.7   | 5.0   | 4.8  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 89.2  | 89.2  | 90.3  | 90.0  | 89.6  | 89.5  | 89.9  | 89.9  | 89.9  | 89.8  | 89.6  | 89.2 |
|                                                                                  |             | MCWB   | 79.5  | 79.2  | 79.2  | 80.5  | 81.6  | 81.7  | 81.8  | 82.0  | 82.1  | 82.3  | 81.9  | 81.1 |
|                                                                                  | 2%          | DB     | 87.4  | 87.4  | 88.2  | 88.2  | 88.4  | 88.2  | 88.5  | 88.6  | 88.6  | 88.7  | 88.2  | 87.7 |
|                                                                                  |             | MCWB   | 79.3  | 79.1  | 79.5  | 80.2  | 81.0  | 80.8  | 80.8  | 81.1  | 81.1  | 81.3  | 80.7  | 80.1 |
|                                                                                  | 5%          | DB     | 86.4  | 86.4  | 87.0  | 87.4  | 87.7  | 87.6  | 87.7  | 87.6  | 87.7  | 87.8  | 87.4  | 86.9 |
|                                                                                  |             | MCWB   | 78.8  | 78.6  | 79.2  | 79.9  | 80.7  | 80.4  | 80.4  | 80.3  | 80.6  | 80.5  | 80.2  | 79.7 |
|                                                                                  | 10%         | DB     | 85.6  | 85.6  | 86.1  | 86.6  | 86.8  | 86.7  | 86.7  | 86.7  | 86.8  | 86.9  | 86.6  | 86.1 |
|                                                                                  |             | MCWB   | 78.5  | 78.3  | 78.8  | 79.6  | 80.3  | 80.1  | 80.0  | 79.9  | 80.1  | 80.1  | 79.9  | 79.2 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 83.4  | 83.1  | 83.4  | 83.9  | 84.9  | 84.6  | 84.4  | 84.4  | 84.8  | 84.9  | 84.2  | 83.2 |
|                                                                                  |             | MCDB   | 86.5  | 86.2  | 86.5  | 87.6  | 88.2  | 88.0  | 88.2  | 88.6  | 88.6  | 88.6  | 88.0  | 87.1 |
|                                                                                  | 2%          | WB     | 81.0  | 80.9  | 81.3  | 81.9  | 82.5  | 82.3  | 81.9  | 82.3  | 82.7  | 82.6  | 82.1  | 81.6 |
|                                                                                  |             | MCDB   | 85.2  | 85.1  | 85.7  | 86.2  | 86.9  | 86.7  | 86.8  | 87.2  | 87.3  | 87.4  | 86.7  | 85.9 |
|                                                                                  | 5%          | WB     | 80.0  | 79.9  | 80.3  | 80.9  | 81.4  | 81.2  | 81.0  | 81.1  | 81.4  | 81.4  | 81.0  | 80.7 |
|                                                                                  |             | MCDB   | 84.5  | 84.4  | 85.1  | 85.6  | 86.1  | 85.9  | 86.1  | 86.2  | 86.3  | 86.3  | 85.8  | 85.1 |
|                                                                                  | 10%         | WB     | 79.2  | 79.1  | 79.5  | 80.1  | 80.6  | 80.4  | 80.2  | 80.2  | 80.5  | 80.5  | 80.3  | 79.9 |
|                                                                                  |             | MCDB   | 83.8  | 83.8  | 84.3  | 84.9  | 85.4  | 85.2  | 85.3  | 85.3  | 85.4  | 85.4  | 85.0  | 84.4 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 6.8   | 7.1   | 7.4   | 7.5   | 8.1   | 8.4   | 8.9   | 9.0   | 9.1   | 9.1   | 8.2   | 7.3  |
|                                                                                  |             | MCDBR  | 7.6   | 7.8   | 8.2   | 8.2   | 8.7   | 9.0   | 9.8   | 10.1  | 10.0  | 9.9   | 9.0   | 8.0  |
|                                                                                  |             | MCWBR  | 4.8   | 4.9   | 5.1   | 5.1   | 5.3   | 5.4   | 5.7   | 5.7   | 5.8   | 5.6   | 5.2   | 4.8  |
|                                                                                  | 5% WB       | MCDBR  | 7.3   | 7.6   | 7.8   | 7.9   | 8.5   | 8.8   | 9.4   | 9.8   | 9.8   | 9.5   | 8.7   | 7.9  |
|                                                                                  |             | MCWBR  | 5.2   | 5.3   | 5.5   | 5.4   | 5.8   | 5.8   | 5.8   | 5.9   | 6.3   | 6.1   | 5.5   | 5.1  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.452  | 0.454 | 0.452 | 0.441 | 0.419 | 0.401 | 0.387 | 0.381 | 0.379 | 0.392 | 0.423 | 0.465 |      |
|                                                                                  | taud        | 2.370  | 2.358 | 2.393 | 2.456 | 2.543 | 2.602 | 2.651 | 2.667 | 2.680 | 2.608 | 2.511 | 2.327 |      |
|                                                                                  | Ebn at Noon | 274    | 277   | 278   | 276   | 277   | 279   | 283   | 289   | 294   | 292   | 281   | 268   |      |
|                                                                                  | Edh at Noon | 39     | 41    | 40    | 37    | 33    | 30    | 29    | 30    | 30    | 31    | 34    | 41    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1570   | 1708  | 1809  | 1779  | 1702  | 1639  | 1679  | 1676  | 1685  | 1676  | 1567  | 1520  |      |
|                                                                                  | RadStd      | 98     | 164   | 95    | 127   | 104   | 125   | 136   | 157   | 111   | 92    | 115   | 102   |      |

**Historical Trends**

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | N/A     | +0.74  | +0.68   | +0.92 | +1.26 | N/A         | N/A   | N/A   | N/A   |
| Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |